

A Packing of 7 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 7 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.799799. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.742632, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 7 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.799799 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 7$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.742632, obtained by deleting 1 of the 8 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 7/6 = 2.800000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N7.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
699949986565025414613916626937883025797774988418299265846276827114392304
868498180147743646500707361382955457950010460784105490952276043214681712
673513590812376606446115418690485716211935099079
```

denominator:

```
250000000000204721494097948950121164440113780049267226074514795625155272
229257818427854215628914854321333301353085880824642199946510005647255227
343743733223590719845854635293674391662124519309
```

B Coordinates

7 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N7.txt`.

```
-250246551930 72124163057 -179171422046 -6336927718085408296 -145087553456 -296450897593 -2962572570426507045840 211319372772 -497565322941 89938603269
-391673063395 556069479280 -326306585586 3063126040004969746838 295749723242 116783695907 26892273732544315832824 -204979512173 -389593223043 -304343158154
-20156545244 -190194910634 624075150306 -4599896626231915772008 -364319459098 75655602888 74158545220210504063752 447319910129 100471881058 -17025774597
-500195956238 -32767424504 253217460847 -26860414225358221309 -510032055610 167902084027 -20558054587872302991325 404137486361 34727264219 -387678927834
-407198034456 -21103585332 428418063714 2533386118796620371895 -484908123593 50864689993 11967429866183124431148 188516443698 242831251955 -198673645713
7118522434 -305036895630 -170801314493 -44455597639507650089841 -38548507358 217282366360 -21521791584859147070 124913922920 -560643944008 32913033051
413210810421 410896097182 353770548424 -5151000411204449248545 326386528042 329819017682 -7692746794697815657757 50638060075 -104167694222 706282263664
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189135468003 -212039737270 29296372514 123849724211-30569675973 304064218105 326068201617 164430635812-97668189820 402951261346 -650113101010 -154279121063
325371716656 482156812641 134083366847 -343881456338491349096 -60095248086 648942650487 118487657984 166628035819 349117432490 57490331728 193901811614
-379685934569 -111544386656 249896456202 1102474123726333905860 238257958275 -425465047783 -654033389512165705693 -251009368355 -38413334601 -461600362254
132145934978 369476938259 190163704280 538759712801-468618486821 361747464503 -263129038390 -11847860414816057144 430171082987 346777435936 152240293570
-2915540872 531783167950 72458782402 127957732971 -39740231706 -355200041518 -128496857922 -31863529423289765107 159388520570 149611138563 146652005959
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594427670351 -6371134022 -314483749150 59798750607 -335398495297 -263139329987 -257404035430 -3983781272857454261 62995094689 54584964743 446426810435
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-108979302729 398773369120 -143498516709 24951872153807495396552 -473676674681 -26985683460 51169634789153251393540 211757872637 88445861738 412735164455
-354354499884 -255419194877 -668320212995 190190578338011672878 -348137637823 -55288499593 -228878873869
68922088309 38722948117 485708382832 245279986784 -293433942311 620649536240 131554174541 12498359280
```